



Web Dashboard — User Guide



Gateway for the Hoymiles HMS-800W-2T solar inverter

Firmware **v0.5.2** and later · Document date **2026-08-11**
github.com/danielguedel/HMS-GW-S3

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1. Introduction

The HMS-GW-S3 gateway reads live data from your Hoymiles HMS-800W-2T solar inverter and makes it available through a built-in **web dashboard** — no app, no cloud account, no installation required. Just open a browser and point it at the gateway's IP address.

This guide covers everything you can do from that dashboard: monitoring live solar output, controlling the relay and auxiliary IOs, changing WiFi/MQTT/network settings, and updating the firmware.

ALSO AVAILABLE

A companion Progressive Web App for viewing and controlling the gateway from outside your home network is covered briefly in §7 and described in full in the project's README.

What you need

- A device with a modern web browser (phone, tablet, laptop, desktop — any OS)
- The gateway connected to your WiFi network (see §2 if this is a first-time setup)
- The gateway's IP address, or its factory-default access point if it isn't on your network yet

2. Getting Started

2.1 First connection (factory-new device)

Out of the box, the gateway cannot yet know your home WiFi credentials, so it opens its own temporary access point:

- 1 On your phone or laptop, open the WiFi settings and connect to the network named `HMS-GW-S3`. No password is required.
- 2 Open a browser and go to `http://192.168.4.1`.
- 3 The dashboard loads. Switch to the **Config** tab (see §5) and enter your home WiFi network name (SSID) and password under the "WiFi" section.
- 4 Click **Save**. The gateway restarts and connects to your home network.

NOTE

Once connected to your home network, the temporary `HMS-GW-S3` access point disappears — you'll need the gateway's new IP address to reach it again (see below).

2.2 Finding the gateway on your network

After the WiFi setup above, the gateway gets its address from your router (DHCP) unless you configured a static IP. The most reliable way to find it:

- Open your router's admin page and look at the list of connected devices for one named `HMS-GW-S3` (or similar).
- Alternatively, check the gateway's status LED behaviour or serial console output if you have access to it (see the project's technical specification for LED colour meanings).

There is no `.local` hostname — the gateway must be reached by its numeric IP address (e.g. `http://192.168.1.42`).

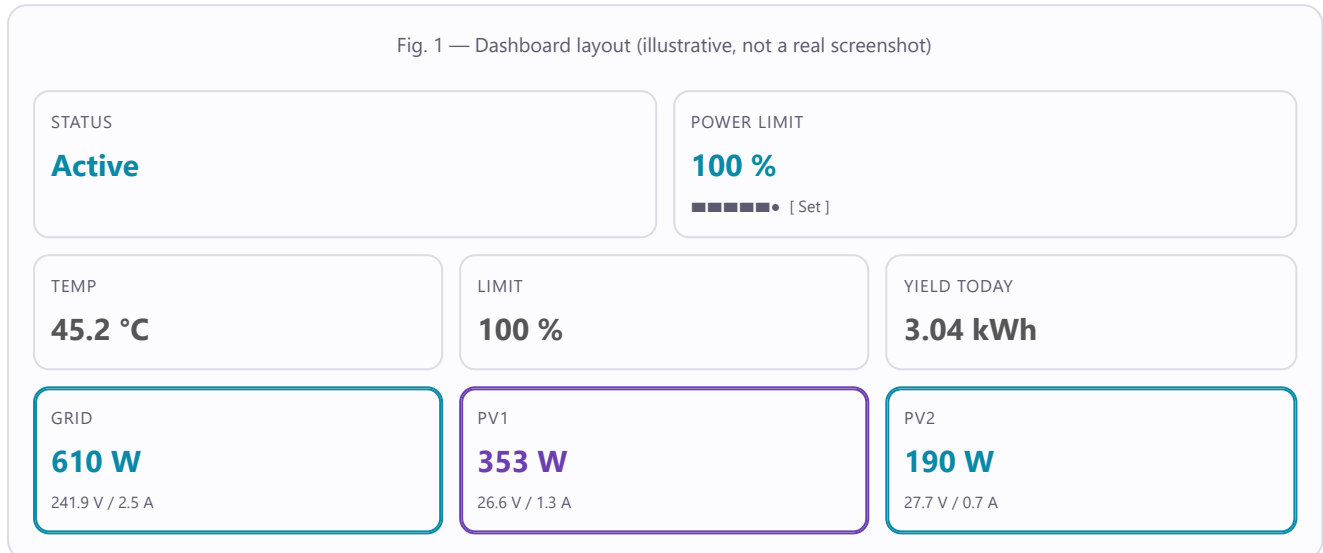
2.3 Bookmark it

Once you know the address, bookmark it in your browser for quick access later. If you later assign the gateway a static IP (Config tab → Network), that address won't change again.

3. The Dashboard Tab

The Dashboard is the default view and shows live data at a glance. It refreshes automatically every 10 seconds — no need to reload the page.

3.1 Layout overview



3.2 Status card

Shows one of four states, colour-coded (see §9 for the full colour legend):

LABEL	MEANING
Active	WiFi, DTU and MQTT are all connected, and the inverter is currently feeding power into the grid. Shown in cyan.
Standby	Everything is connected, but the inverter isn't producing right now (e.g. at night, or too little sunlight).
No data	Everything is connected, but no reading has arrived yet (briefly shown right after startup).
Offline	WiFi, the DTU link, or MQTT is not connected.

3.3 Power Limit card

Lets you cap how much power the inverter is allowed to feed into the grid, as a percentage of its rated output.

- 1 Drag the slider to the desired percentage.
- 2 Click **Set** to send the new value.
- 3 The card border turns **pink** while the change is in flight, and back to **cyan** once the inverter confirms it (usually within a few seconds).

TIP

If a sent limit never turns cyan, the inverter may be unreachable (e.g. the DTU link is down) — check the Status card and the DTU settings in the Config tab.

3.4 Temp / Limit / Yield today / Total cards

Inverter temperature, the currently confirmed power limit, and today's/all-time energy yield across both PV strings and the grid connection combined.

3.5 Grid / PV1 / PV2 cards

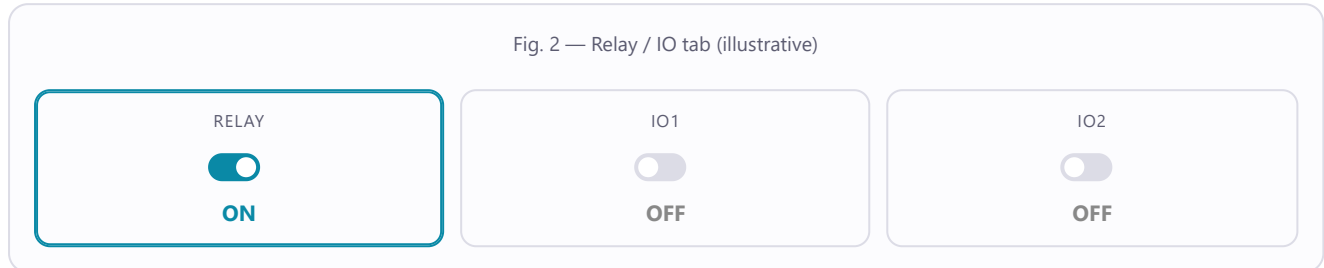
Live power (W), voltage and current for the grid connection and each of the two PV strings. The card border and the power figure change colour depending on the current output level — see §9 for exact thresholds. This gives you an at-a-glance sense of how much each string is producing, even from across the room.

3.6 Header

At the top of every page: three status dots (WiFi / DTU / MQTT — lit when connected), a language toggle (DE / EN), and an info bar showing the device's MAC address, firmware version and build number, plus the current date and time.

4. The Relay / IO Tab

Four independent switchable channels: the **Relay** output and three general-purpose **IO** pins (which channel maps to which physical GPIO depends on your configuration — see Config → System/GPIO in the technical specification for wiring details).



- 1 Tap or click a toggle switch to turn that channel on or off.
- 2 The switch and the small ON/OFF label update immediately once the gateway confirms the change.
- 3 A card whose channel is currently **ON** gets a glowing cyan border, so active channels stand out even in a quick glance.

Typical uses: switching an external contactor or pump via the relay, or reading/driving auxiliary equipment via the IO pins (exact wiring and pin assignment is a one-time setup done by whoever installed the gateway).